

Schur-Complement Hessian with Two-Timescale Tracking

A Short Note on Possible Second-Order Extensions of TTGDA

Feier Lyu

June 2026

Scope. This note is an exploratory technical note motivated by Lin–Jin–Jordan’s work on two-timescale gradient descent ascent (TTGDA) for nonconvex-concave minimax optimization. It does not claim that the TTGDA theorem is incomplete or incorrect, nor does it claim that second-order minimax optimization is new. The goal is more modest: to isolate a proof-level question about whether the first-order two-timescale tracking mechanism can also support a second-order saddle-escaping analysis for the value function.

Motivation. Consider the minimax problem

$$\min_{x \in \mathbb{R}^d} \max_{y \in \mathcal{Y}} f(x, y),$$

with the value function

$$\Phi(x) := \max_{y \in \mathcal{Y}} f(x, y).$$

In the smooth nonconvex-strongly-concave setting, where f is smooth and strongly concave in the y -variable, TTGDA can be viewed as an inexact gradient method on Φ . If the inner maximizer

$$y^*(x) \in \arg \max_{y \in \mathcal{Y}} f(x, y)$$

is available exactly, then under suitable regularity assumptions,

$$\nabla \Phi(x) = \nabla_x f(x, y^*(x)).$$

TTGDA does not solve the inner maximization to high accuracy at every outer step. Instead, it uses the current tracker y_t , so the x -update uses

$$\nabla_x f(x_t, y_t)$$

rather than

$$\nabla \Phi(x_t) = \nabla_x f(x_t, y^*(x_t)).$$

Thus the first-order analysis controls the gradient error through the tracking error between y_t and $y^*(x_t)$. The question in this note is whether this tracking viewpoint can be pushed one derivative higher: can it also control the curvature of Φ well enough to support a perturbed-gradient-descent-style saddle-escaping argument?

A small check: first-order stationarity is not enough. First-order stationarity of Φ does not rule out negative curvature. Consider

$$f(x, y) = \cos x + a \sin x y - \frac{1}{2} y^2, \quad x \in \mathbb{R}, \quad y \in [-2, 2],$$

where $0 < a < 1$. For every fixed x , the function is 1-strongly concave in y , and

$$\partial_y f(x, y) = a \sin x - y.$$

Hence the maximizer is

$$y^*(x) = a \sin x,$$

which lies in the interior of $[-2, 2]$. Therefore

$$\Phi(x) = f(x, y^*(x)) = \cos x + \frac{1}{2}a^2 \sin^2 x.$$

Differentiating gives

$$\Phi'(x) = -\sin x + a^2 \sin x \cos x, \quad \Phi''(x) = -\cos x + a^2(\cos^2 x - \sin^2 x).$$

At $x = 0$,

$$\Phi'(0) = 0, \quad \Phi''(0) = -1 + a^2 < 0.$$

Thus $x = 0$ is first-order stationary for the value function but not second-order stationary for the outer minimization problem. This is not a counterexample to TTGDA; it only shows that a first-order guarantee for Φ does not by itself exclude negative-curvature behavior.

Effective curvature of the value function. The natural second-order object is the Hessian of the value function. In this paragraph, I use additional regularity assumptions beyond the first-order TTGDA result: assume for simplicity that $y^*(x)$ is an interior maximizer, that f is sufficiently smooth, and that $f_{yy}(x, y^*(x))$ is negative definite. From the first-order condition

$$\nabla_y f(x, y^*(x)) = 0,$$

differentiating with respect to x gives

$$f_{yx} + f_{yy} Dy^*(x) = 0, \quad Dy^*(x) = -f_{yy}^{-1} f_{yx},$$

where all derivative blocks are evaluated at $(x, y^*(x))$. Since

$$\nabla \Phi(x) = \nabla_x f(x, y^*(x)),$$

we obtain the Schur-complement Hessian

$$\nabla^2 \Phi(x) = f_{xx} - f_{xy} f_{yy}^{-1} f_{yx}.$$

This formula should be understood in the clean interior-maximizer case. If $y^*(x)$ lies on the boundary of $\mathcal{Y}$, or if the active constraint set changes, then the simple Schur-complement formula may need to be replaced by a KKT or active-set version.

For the toy example above, at $(x, y) = (0, 0)$,

$$f_{xx} = -1, \quad f_{xy} = a, \quad f_{yy} = -1.$$

Hence

$$f_{xx} - f_{xy} f_{yy}^{-1} f_{yx} = -1 + a^2 < 0,$$

which matches $\Phi''(0)$. For example, when $a = 1/2$, the exact curvature is -0.75 , and a central finite-difference check with $h = 10^{-3}$ agrees to the displayed precision.

Diagnostic curvature along the TTGDA trajectory. At the actual TTGDA iterate (x_t, y_t) , a natural diagnostic object is

$$\hat{H}_t := f_{xx}(x_t, y_t) - f_{xy}(x_t, y_t)f_{yy}(x_t, y_t)^{-1}f_{yx}(x_t, y_t).$$

If $\lambda_{\min}(\hat{H}_t)$ is sufficiently negative, one might hope that the true value-function Hessian $\nabla^2\Phi(x_t)$ also has a negative direction. A local Lipschitz-type bound for the Schur-complement map may be obtainable under uniform strong concavity in y and Hessian-Lipschitz assumptions. For example, one would expect a perturbation estimate of the form

$$\left\| \hat{H}_t - \nabla^2\Phi(x_t) \right\| \leq C \|y_t - y^*(x_t)\|,$$

where C depends on smoothness bounds, the Hessian-Lipschitz constant, and the strong concavity parameter.

This local bound alone would not yet give a saddle-escaping theorem. The harder issue is timing: the curvature estimate must be accurate at an iterate where the outer gradient is already small and the reduced landscape has negative curvature. The first-order TTGDA analysis controls tracking-related error terms sufficiently for a descent argument, but a second-order argument may require stronger pointwise or high-probability control at small-gradient iterates.

Perturbations and re-tracking. There are two possible roles for $\hat{H}_t$. First, it could be used only as a diagnostic or certification object in the analysis, while the algorithm still uses random perturbations in x , in the spirit of perturbed gradient descent. Second, one could explicitly compute and follow a negative-curvature direction of $\hat{H}_t$. The latter would no longer be vanilla perturbed gradient descent; it would be closer to a Hessian-vector or explicit second-order method within the two-timescale framework. The diagnostic route raises a concrete tracking question. If a random perturbation is added to the x -variable, then the old y_t may no longer be a good tracker of $y^*(x_t)$ at the perturbed point. Thus the fast y -recursion would need to re-track the new maximizer quickly enough before the outer x -updates lose the negative-curvature information needed for the escape argument. In this sense, the main difficulty is not the Schur-complement formula itself, but the loss and recovery of tracking accuracy after perturbation.

Main question. Can the first-order TTGDA tracking analysis be upgraded toward a second-order saddle-escaping guarantee within essentially the same two-timescale framework? More concretely, if $\hat{H}_t$ is used to detect or certify negative curvature of the reduced landscape, can the tracking error before and after a perturbation in x be controlled tightly enough to exploit negative curvature of the true value function Φ ?

References. Tianyi Lin, Chi Jin, and Michael I. Jordan. *Two-Timescale Gradient Descent Ascent Algorithms for Nonconvex Minimax Optimization*. Journal of Machine Learning Research, 26(11):1–45, 2025.

Chi Jin, Rong Ge, Praneeth Netrapalli, Sham M. Kakade, and Michael I. Jordan. *How to Escape Saddle Points Efficiently*. ICML, 2017.

Chi Jin, Praneeth Netrapalli, and Michael I. Jordan. *What is Local Optimality in Nonconvex-Nonconcave Minimax Optimization?* ICML, 2020.

Luo Luo, Yujun Li, and Cheng Chen. *Finding Second-Order Stationary Points in Nonconvex-Strongly-Concave Minimax Optimization*. NeurIPS, 2022.

Jiseok Chae, Kyuwon Kim, and Donghwan Kim. *Two-timescale Extragradient for Finding Local Minimax Points*. ICLR, 2024.

Ziyi Chen, Zhengyang Hu, Qunwei Li, Zhe Wang, and Yi Zhou. *A Cubic Regularization Approach for Finding Local Minimax Points in Nonconvex Minimax Optimization*. arXiv:2110.07098, 2021.